Therefore the separation distance between the isotopes is $\Delta d = d_{238} - d_{235} = 1.2\,\text{cm}$

(b) The time taken by each isotope to complete one semi-circular path are

$$t_{235} = \frac{\text{magnitude of the displacement}}{\text{velocity}}$$

$$= \frac{97.6\times10^{-2}}{1.00\times10^{5}} = 9.76\times10^{-6}\,\text{s} = 9.76\,\mu\text{s}$$

$$t_{238} = \frac{\text{magnitude of the displacement}}{\text{velocity}}$$

$$= \frac{98.8\times10^{-2}}{1.00\times10^{5}} = 9.88\times10^{-6}\,\text{s} = 9.88\,\mu\text{s}$$

Note that even though the difference between mass of two isotopes are very small, this arrangement helps us to convert this small difference into an easily measurable distance of separation. This arrangement is known as mass spectrometer. A mass spectrometer is used in many areas in sciences, especially in medicine, in space science, in geology etc. For example, in medicine, anaesthesiologists use it to measure the respiratory gases and biologist use it to determine the reaction mechanisms in photosynthesis.

3.10.3 Motion of a charged particle under crossed electric and magnetic field (velocity selector)

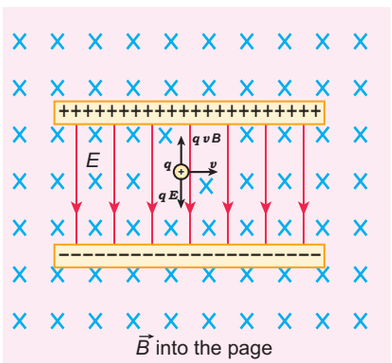


Figure 3.48 Velocity selector

Let us consider an experimental arrangement to illustrate velocity selector as shown in Figure 3.48. In the region of space between the parallel plates of a capacitor which produce uniform electric field $\vec{E}$, a uniform magnetic field $\vec{B}$ is maintained perpendicular to the direction of electric field.

Suppose a charged particle with charge q enters the space from left side with a velocity $\vec{v}$, the net force on the particle is

$$\vec{F} = q\left(\vec{E} + \vec{v}\times\vec{B}\right)$$

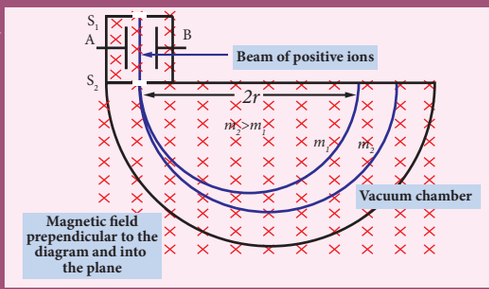
For a positive charge, the electric force on the charge acts in downward direction whereas the Lorentz force acts upwards. When these two forces balance each other, then

$$qE = qvB$$

$$\Rightarrow v = \frac{E}{B} \qquad (3.62)$$

Note This principle is used in Bainbridge mass spectrograph to separate the isotopes. This concept is explained in Example (3.21).

Schematic diagram of Bainbridge mass spectrometer



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